End-to-End Design of Vision-Based Actuator Recognition and Proactive Routine Integration on LG CLOiD under Constrained Operation

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***Abstract*—Many homes still have wall switches, push buttons and dials that are not connected to any network. Smart-home routines cannot use them unless someone registers each device by hand. This paper designs the whole process of fixing that, from finding a control with the vision module of LG CLOiD to running it inside a proactive routine. The process has seven stages in three phases: setup, runtime and learning. Public information on CLOiD is thin, so we do not know how far the robot can actually handle these tasks, and we do not assume it works well. We treat it as deliberately limited and define four capability levels, L0 to L3. The system first measures what the robot can do, then turns on stages only as far as that measurement allows. Wireless control comes first and physical manipulation is the fallback; a risk-based autonomy policy and a safety check guard every action. We give the process, architecture, data model, requirements and a plan to demonstrate three scenarios end to end. Implementation and measurements come later. Every number in this paper is a target, not a result.**

***Index Terms*—smart home, service robot, vision-language model, vision-language-action, proactive routine, actuator recognition, end-to-end system**

I. Introduction

A. Background and Problem

Routines in current smart-home platforms follow simple rules: when a time or sensor condition holds, control the registered devices. Two things limit this. First, legacy controls such as wall switches, push buttons, toggles and dials have no network interface, so a routine cannot target them. Second, the user has to register every device and its control method, so automation grows only as fast as the user's patience.

Home humanoids such as LG's CLOiD, shown at CES 2026, make it realistic to place a robot with cameras and arms in a living room [1]. We could not find an official specification, though. Public coverage is mostly demo-level, so we do not know how far it can actually handle tasks in a real home: how reliable its vision, navigation and manipulation are.

B. Goal and Contributions

Our goal is to show the whole process in one working system: CLOiD's vision module finds a control, the system works out what it operates, the control is added to a routine, and later the routine runs it, with a fallback when things fail. The paper contributes four things.

1) **An end-to-end process.** Seven stages in three phases, each with defined inputs, outputs, failure handling and the capability level it needs (Sections IV–VII).
2) **A constrained-operation model.** We assume as little as possible about the robot and open up stages in four capability levels, L0 to L3 (Section III).
3) **Mapping by acting and observing.** A mapping from control to device is accepted only after the system sees the effect (Section V).
4) **A demonstration plan.** Three scenarios that together cover every stage, with metrics for the whole run and for each stage (Section X).

C. Scope

This is the Phase 2 deliverable of our software engineering course. It covers the process, architecture, requirements and a verification plan. It has no implementation or measured results. Reverse engineering the vendor's internal APIs and modifying the robot hardware are out of scope.

II. Related Work

Grounding language in action. SayCan [2] filters a language model's plan by what the robot can actually do. RT-2 [3] and OpenVLA [4] are vision-language-action (VLA) models that go from perception to control in one model. We treat the VLA planning step as a replaceable module.

Open-vocabulary detection. OWL-ViT [5] and Grounding DINO [6] detect object categories from a text query without retraining. Switches, buttons and dials vary a lot in shape, so this is a practical way to get candidates.

TABLE I
CAPABILITY LEVELS

Level	Robot function added	What the system can do
L0	Receive the video stream (observe only)	Detect controls; watch the user operate them and propose candidate mappings
L1	Move or change posture; voice or screen notices	Re-observe from a better view; ask the user to confirm; notify
L2	Wireless commands (IR/Wi-Fi/BT) through a hub, with CV state check	Control devices in a closed loop; act proactively on low-risk devices
L3	Physical manipulation with arms and hands	Fall back to physical operation when wireless control is missing or fails

Smart-home interoperability. Standards such as Matter [7] cover devices that are already on the network. We look at the controls that stay outside such standards.

Difference. Prior work mostly executes a given task. We cover the whole loop: discovering a control, mapping it, putting it in a routine, running it, falling back when it fails, and learning from the result, all while the robot's real capability is unknown.

III. CONSTRAINED OPERATION MODEL

A. What We Know and Assume

Public reports describe CLOiD as having a head with cameras and sensors, two seven-degree-of-freedom arms with five-finger hands, and a wheeled base [1]. We could not confirm battery life, arm reach, button-press accuracy or external APIs. Since we do not know how far the robot can handle a task, the system has to find that out itself before relying on it. The design therefore follows three rules.

- **Assume little.** The only thing we count on is a video stream. Everything else is optional and used if available.
- **Hide the robot behind an interface.** All robot functions go through a Robot Abstraction Layer (RAL), so the real CLOiD, a simulator or a webcam stand-in can be swapped in.
- **Measure, do not trust.** A Capability Probe measures what the robot can do, and the measured result sets the level.

B. Capability Levels L0–L3

Table I defines the levels. Each level includes the one below it. If a measurement falls under its threshold, the system drops one level at once.

C. Capability Probe

At start-up and periodically, the Capability Probe measures the following.

- **L0:** frame rate, latency and resolution; the distance at which a test marker is recognized.
- **L1:** success rate of moving to a given spot and observing again.
- **L2:** accuracy of the CV state check after commanding a test smart plug.
- **L3:** success rate of N press attempts on a calibration button panel.

The pass thresholds (for example, 9 of $N{=}10$ presses for L3) are first guesses that we will tune in the test environment.

IV. END-TO-END PROCESS

Fig. 1 shows the whole process, and Table II lists what each stage takes in and gives out. The process has three phases.

- **Phase 1, Setup (S1–S3).** Find a control, map it to a device, verify the mapping, and turn it into a routine with the user's approval. This phase runs when a new control appears.
- **Phase 2, Runtime (S4–S6).** A trigger fires, the system infers the intent, sends a wireless command and checks the result with CV. If the check fails, the robot operates the control physically, but only after a safety check.
- **Phase 3, Learning (S7).** Every run, including cancelled ones, updates the channel priorities and mapping confidence that Phases 1 and 2 use.

V. PHASE 1: SETUP

A. S1: Recognize Controls

Actuator Perception works in three steps.

1) **Candidate extraction.** An open-vocabulary detector [5], [6] gets text queries such as "wall switch," "push button," "toggle," "knob" and "remote control," and returns bounding boxes.
2) **Classification.** The crop of each candidate goes to a VLM, which returns structured output: control type (toggle, push, dial, remote), visible state (on or off), and possible actions (press, turn).
3) **Localization.** Depth data gives a 3D position in the room frame, and the control is stored as an *Actuator* record. An appearance embedding is stored too, so the same control can be re-identified later.

B. S2: Map and Verify

A mapping between a control and its target device is built in two ways.

(a) Passive observation (L0 and up). When the user operates a control and a nearby device changes state, the system records a candidate mapping. This works even if the robot cannot move.

(b) Actuate and observe (L2/L3). For a candidate mapping, the system (1) captures the target device region before acting, (2) operates the control through its channel (a physical press at L3), and (3) compares the region afterward. If the change passes a threshold and the VLM agrees, the mapping becomes verified.

If verification fails, the control method is registered again, and after a set number of retries the system asks the user. Fig. 2 shows the life cycle of a mapping record.

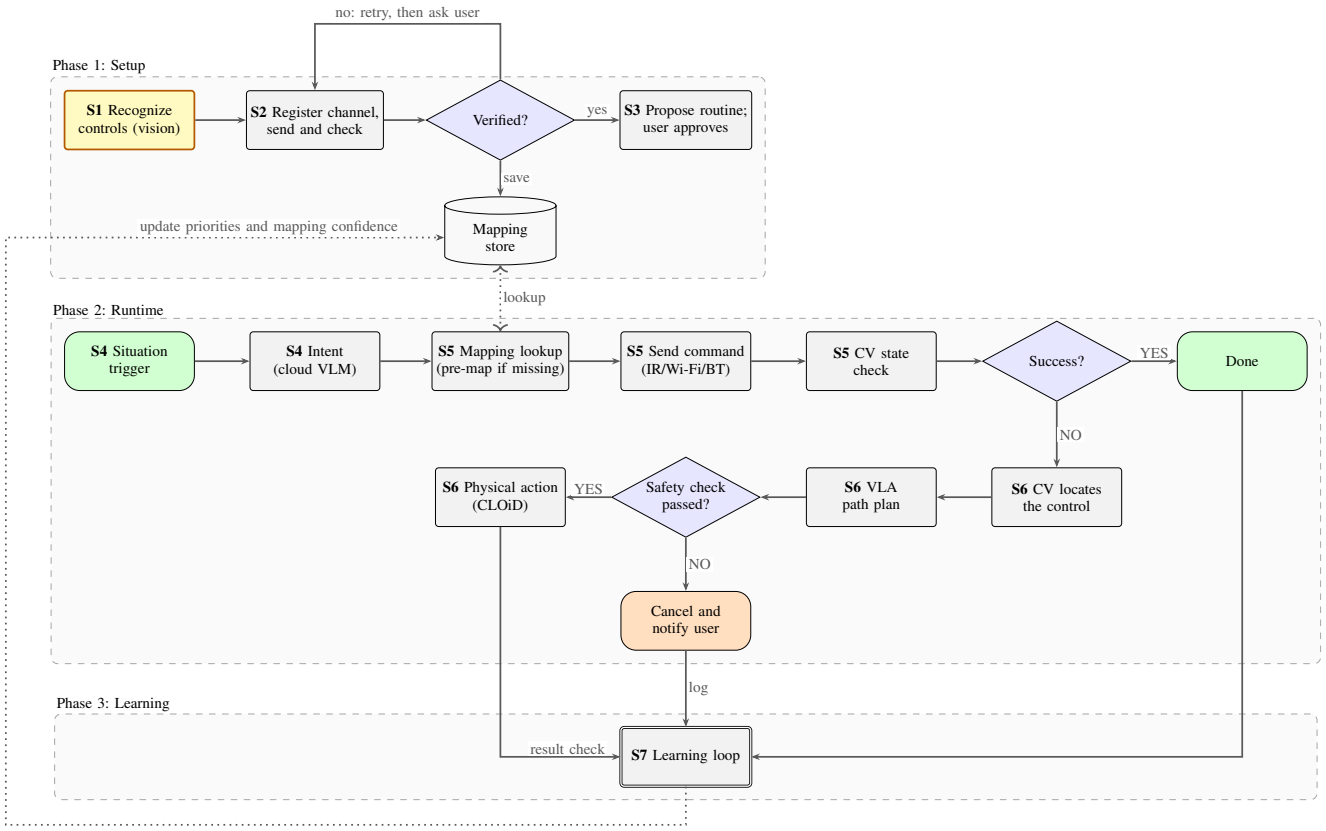


Fig. 1. End-to-end process. Yellow marks the new step (vision-based control recognition). Wireless control comes first, physical action is a fallback after a failed CV check, and a failed safety check cancels it. Every path ends in the learning loop.

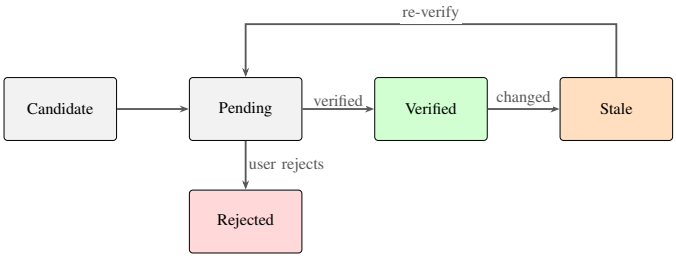


Fig. 2. Life cycle of a mapping record. If the look or position of a control changes, the mapping goes stale and is checked again.

C. S3: Integrate into Routines

A verified mapping enters routines in two ways. The system can watch for repeated user behavior, such as switching on the living-room light at the same time each day, and *propose a routine*. The user can also pick a verified control and build a routine by hand. Either way, a routine is added only after the user approves. An approved routine becomes a trigger source for Phase 2.

VI. Phase 2: Runtime

A. S4: Trigger and Intent

A sensor, time or routine event starts the run. The Routine Engine sends the context and a camera frame (crops only) to the cloud VLM, which returns the intent and the target device. Confidence for acting on it is a weighted sum, as in (1).

$$c = \lambda_1 \, c_{\text{vlm}} + \lambda_2 \, c_{\text{ctx}} + \lambda_3 \, c_{\text{hist}} \tag{1}$$

c_{vlm} is the VLM's confidence in the intent, c_{ctx} is the reliability of the trigger, and c_{hist} is the past acceptance rate. What happens next depends on the risk class of the device (Table III).

B. S5: Wireless Control and CV Check

The Orchestrator applies the capability gate first. It then looks up the mapping (if there is none, it pre-maps the device) and sorts the device's wireless channels by the score in (2).

$$s_{d,k} = w_1 \, \hat{p}_{d,k} - w_2 \, \hat{t}_{d,k} - w_3 \, c_k \tag{2}$$

$\hat{p}$ is the estimated success rate, $\hat{t}$ is normalized latency and c_k is the intrusiveness cost of the channel. The physical (PHYS) channel has the highest cost and is a candidate *only when there is no wireless channel or all of them failed*. The Orchestrator sends the command, then asks State Verifier to confirm the expected change with CV. On success, the run goes straight to S7. If every wireless channel fails and the level is L3 or above, it goes to S6; below L3, it notifies the user.

C. S6: Physical Fallback

CV locates the control, a VLA model plans the approach and the press, and the Safety Guard checks three things before

TABLE II
STAGE SUMMARY: INPUTS, OUTPUTS, FAILURE HANDLING AND MINIMUM CAPABILITY LEVEL

Stage	Input	Output	Component	On failure	Min. level
S1 Recognize controls	Video frames	Actuator records (type, pose, embedding)	Actuator Perception	Re-observe (L1) or skip the candidate	L0
S2 Map and verify	Actuator records; user actions; channel list	Verified mapping	Mapping Builder, State Verifier	Retry a set number of times, then ask the user	L0 passive, L2 active
S3 Integrate into routines	Verified mapping; usage pattern	Approved routine	Routine Engine	Keep as a suggestion	L0
S4 Trigger and intent	Sensor, time or routine event; context	Intent, target device, confidence	Routine Engine, Intent Service	Suggest only, or ask the user	L0
S5 Wireless control and CV check	Intent; mapping; channel scores	Command result; CV-confirmed state	Control Orchestrator, State Verifier	Next channel; then go to S6	L2
S6 Physical fallback	Failed S5; control pose	Physical action; CV-confirmed state	VLA Planner, Safety Guard, Manipulator Adapter	Cancel and notify the user	L3
S7 Learn	Results and logs of S2, S5, S6	Updated scores and confidence	Learning & Priority	Keep the old values	L0

TABLE III
AUTONOMY POLICY BY RISK CLASS

Risk	Examples	Allowed behavior
Low	Lights, fan	High confidence: act, then notify. Medium: ask first. Low: suggest only.
Medium	TV, air purifier	Ask, then act. Low confidence: suggest only.
High	Heaters, power strips	No automatic action and **no physical action**; suggest only.

anything moves: (i) obstacles or people near the path, (ii) the risk class of the target, and (iii) how sure the position estimate is. If any check falls short, the action is cancelled and the user is notified. We will use ISO 13482 [9] as a reference when we detail the safety requirements for a personal care robot. If the checks pass, CLOiD acts and CV checks the result. Emergency stop and a drop to L2 or below are always available.

VII. PHASE 3: LEARNING

A. S7: Learning Loop

Every run writes an *ExecutionLog* entry: stage outcomes, channel, latency and reason. Given a result $y \in \{0, 1\}$, the channel success rate is updated as an exponential moving average, as in (3).

$$\hat{p}_{d,k} \leftarrow (1 - \eta)\, \hat{p}_{d,k} + \eta\, y \tag{3}$$

Routine priority follows the acceptance rate of proposals and how often the user rejects or cancels. If the user keeps rejecting a routine, its autonomy level goes down one step. Mapping confidence is lowered when a later CV check disagrees with the stored mapping, and the mapping goes stale (Fig. 2).

VIII. ARCHITECTURE AND DATA MODEL

A. Architecture

Fig. 3 shows four tiers: cloud, edge (home hub), robot and home. All robot functions pass through the RAL, and the level from the Capability Probe acts as a gate in the Orchestrator. Raw video is processed at the edge or below; the cloud only receives crops and metadata (NFR-05). Frames are not stored by default, regions with people are masked, and proactive features are opt-in.

B. Data Model

Fig. 4 shows the core entities. A *Mapping* links an *Actuator* to a *Device*, and a device can have several *ControlChannel*s. Every execution leaves an *ExecutionLog* entry, which is what makes a run traceable from S1 to S7.

IX. REQUIREMENTS

We follow the writing rules of ISO/IEC/IEEE 29148 [8]: each requirement is single, verifiable and traceable. The **resident** approves or rejects routines and is ultimately responsible. **CLOiD** supplies observation, movement and manipulation. The **cloud VLM** infers intent. The **smart-home hub** sends IR, Wi-Fi and Bluetooth commands. Table IV lists the functional requirements with the stage each one belongs to. Priority is M (must), S (should) or C (could).

The values in Table V are targets that we will adjust in the test environment.

X. DEMONSTRATION AND VERIFICATION PLAN

A. Demo Setup

The demo runs in a mock living room in our lab. The target is a lamp that has a wall switch and can also be driven through a smart plug (Wi-Fi) or an IR hub. If CLOiD is available we use it; if not, we use a RAL stand-in (a webcam, a simulator, or a simple stand-in arm) and keep the process unchanged. The Capability Probe runs first, and its result decides which scenarios can run on the real robot. The exact devices and environment are not fixed yet.

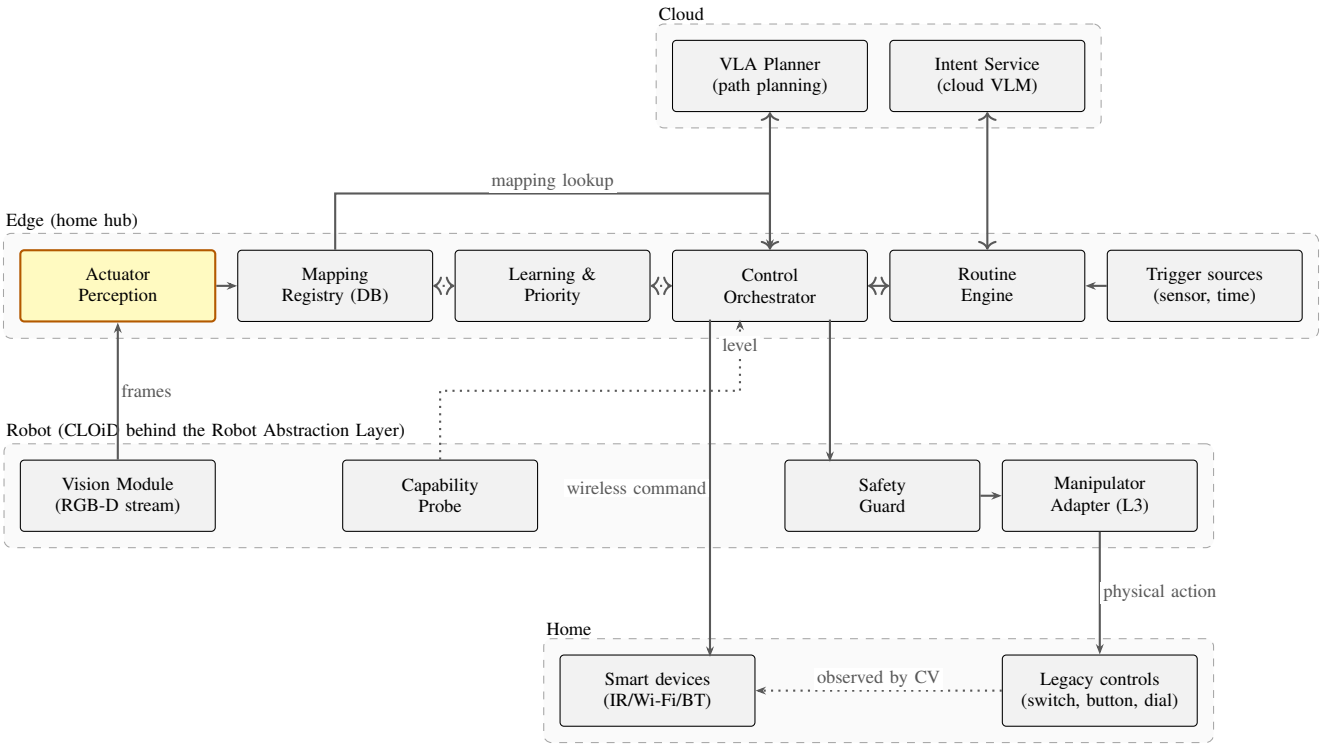


Fig. 3. System architecture. Yellow marks the new component (actuator recognition). Solid lines are control and data flow; dotted lines are reference, update and observation.

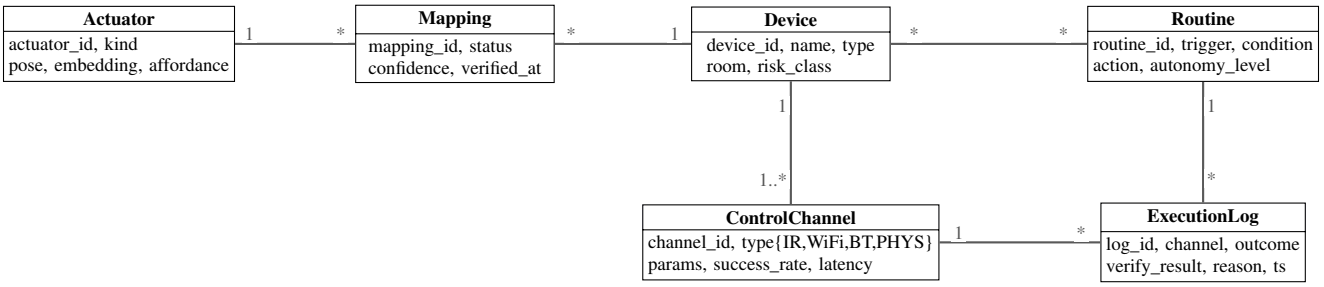


Fig. 4. Core data model (conceptual). A Mapping ties an Actuator to a Device, and every execution leaves an ExecutionLog entry.

B. Scenarios

Table VI lists three scenarios. Together they cover all seven stages, and running Scenario 3 and then Scenario 2 on the same control is the full end-to-end demo.

Fig. 5 shows Scenario 2, the longest runtime path. If the wireless command works, the run ends after step 7 and then goes to S7, which is Scenario 1.

C. Metrics

We report results for the whole run and for each stage (Table VII). A failed run is attributed to the first stage that failed, using the logs from FR-16.

Manual registration is the baseline for E10. Unit and integration tests for each module run in parallel, following the FR traceability table.

D. Risks

We plan four iterations: (1) RAL, mock environment and S1; (2) S2, S3 and the Registry; (3) S4–S6 with the Safety Guard; (4) S7, the full end-to-end runs and evaluation. Each iteration ends with tests for the requirements it covers.

XI. Conclusion

We designed the whole process of bringing legacy controls into proactive routines with CLOiD's vision module: seven stages in three phases, from finding a control to learning from its use. Because we cannot say how well the robot performs, the design uses capability levels L0 to L3 and a Capability Probe to open stages step by step and drop them when needed. Wireless control comes first with a physical fallback, a risk-based policy limits what the system does on its own, and every run is logged stage by stage. Next, we will build the stages in a mock environment, run the three scenarios end to end and

TABLE IV
FUNCTIONAL REQUIREMENTS (FR) AND TRACEABILITY

ID	Requirement	Pri.	Stage	Component	Verification
FR-01	Detect controls (switch, button, dial, remote) in video and classify their type.	M	S1	Actuator Perception	Precision/recall on labeled set
FR-02	Generate candidate mappings between a control and a target device, using spatial context and observed user actions.	M	S2	Mapping Builder	Top-1 mapping accuracy
FR-03	Register a control method (IR, Wi-Fi, BT or physical) per device.	M	S2	Control Orchestrator	Registration scenario test
FR-04	After sending a command, check the expected state change with CV; on failure, ask for re-registration.	M	S2, S5	State Verifier	State-judgment accuracy
FR-05	Store verified mappings durably and look them up before acting.	M	S2, S5	Mapping Registry	Unit tests
FR-06	Receive sensor, time and routine events as situation triggers.	M	S4	Routine Engine	Event-injection test
FR-07	Extract intent and target device with the cloud VLM.	M	S4	Intent Service	Intent accuracy, latency
FR-08	For an unregistered device, guide the user through pre-mapping before acting.	S	S2, S5	Mapping Builder	Scenario test
FR-09	When wireless verification fails, locate the control with CV.	M	S6	Actuator Perception	Localization error
FR-10	Plan an approach path and manipulation with a VLA model.	S (L3)	S6	VLA Planner	Plan success rate
FR-11	Check for obstacles and hazards; if the check fails, cancel and notify the user.	M	S6	Safety Guard	Hazard-injection test
FR-12	After a physical action, verify the result with CV.	M (L3)	S6	Manipulator Adapter	Action success rate
FR-13	Update channel priority and mapping confidence from results.	S	S7	Learning & Priority	Update-rule unit tests
FR-14	Propose a verified mapping as a routine and add it on approval.	M	S3	Routine Engine	Proposal-approval scenario
FR-15	Measure capability with the Capability Probe and gate stages by level.	M	all	Capability Probe	Level-downgrade test
FR-16	Log the outcome and latency of every stage so that a run can be traced end to end.	M	all	All components	End-to-end scenario runs

TABLE V
NON-FUNCTIONAL REQUIREMENTS (NFR, TARGETS)

ID	Requirement
NFR-01	Performance: median time from trigger to a verified wireless command is 5 s or less.
NFR-02	Recognition: control detection precision 0.90 or higher and recall 0.85 or higher in the test room.
NFR-03	Accuracy: confirmed mappings that cause a wrong action are 1% or fewer.
NFR-04	Safety: no physical action after a failed safety check; at L3 or above, motion stops within 1 s of a stop command.
NFR-05	Privacy: no raw video leaves the home; only crops and metadata go out.
NFR-06	Availability: if the robot or cloud fails, drop to a lower level; manual control is never affected.
NFR-07	Extensibility: a new control channel is added as an adapter without changing the core.
NFR-08	Explainability: every automatic action keeps its intent, confidence, channel and reason in a log.
NFR-09	User control: proactive features can be paused at once or switched off one by one.

replace the targets in this paper with measured values. We will test the L3 fallback if we can get access to CLOiD.

TABLE VI
DEMO SCENARIOS AND THE STAGES THEY COVER

No.	Case	Stages	Success criterion
1	Wireless works	S4, S5, S7	Lamp turns on and CV confirms it within 5 s
2	Wireless fails; physical fallback	S4–S7	Switch is pressed, CV confirms; with an obstacle in the path the action is cancelled
3	New control to routine	S1–S3 (then S4–S7)	Mapping verified, routine approved and run once

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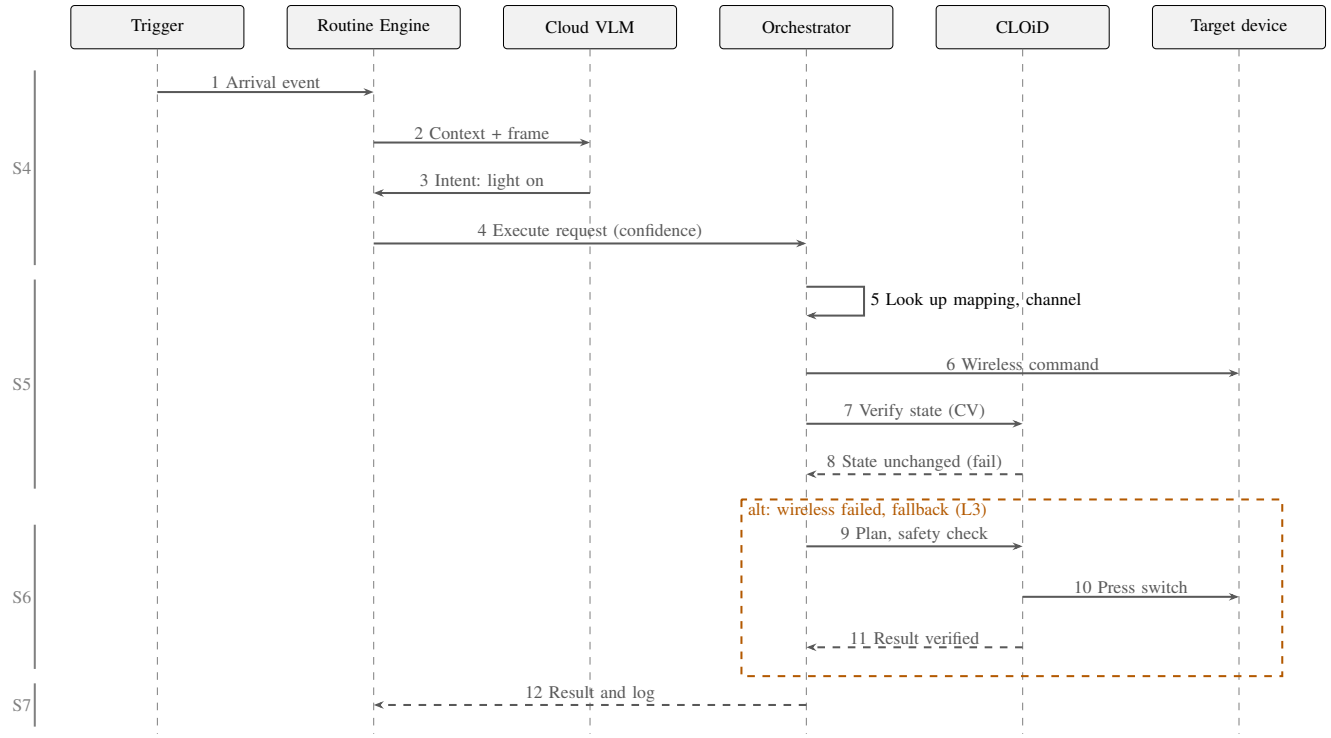


Fig. 5. Sequence for Scenario 2: a failed wireless command and a physical fallback, with stage labels on the left. Dashed arrows are replies or results.

TABLE VII
EVALUATION METRICS AND TARGETS (HYPOTHESES)

Metric	Definition	Target
E1	End-to-end success rate per scenario	90 / 80 / 70% (Sc. 1 / 2 / 3)
E2	Success rate of each stage S1–S7	None below 80%
E3	Latency per stage; trigger-to-verified total	5 s or less (Sc. 1)
E4	Share of runs that reach S6, with causes	Report only
E5	Control detection precision / recall (S1)	0.90 / 0.85
E6	Top-1 accuracy of candidate mappings (S2)	80% or more
E7	CV state-check accuracy (S5)	95% or more
E8	Physical action success rate (S6, L3)	90% or more
E9	Proposal acceptance / false-trigger rate (S3, S4)	70% / under 10%
E10	Time to register a device vs. manual	Shorter than manual

TABLE VIII
MAIN RISKS AND MITIGATION

ID	Risk	Mitigation
R1	No access to a CLOiD API or SDK	RAL with webcam or simulator stand-in; do L0–L2 first
R2	VLM misreads or is slow	Validate structured output; ask the user; cache results
R3	Privacy of home video	Edge processing, crop-only upload, opt-in
R4	Safety of physical action	Safety Guard; no action on high-risk devices; level drop
R5	Device has no known IR code	Learning IR hub; physical fallback
R6	Wrong proactive action	Risk-class policy; lower autonomy after rejections
R7	One failing stage stops the end-to-end demo	Stage-wise logs, retries and fallbacks; a recorded run as backup

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